Problem Solving with General Agents: 90-Minute Crash Course

Mode 1: Problem-Solving, Chapter 2/3 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/problem-solving-crash-course

EN: How to solve problems with a general agent (Claude Code, OpenCode, Cowork, OpenWork) through 7 Principles covering 80% of real-world use. 5 bullets give 60% of the value: Action over talk, Code over prose, Verify don't trust, Small reversible steps, Files are memory.
Roman Urdu: General agent (Claude Code, OpenCode, Cowork, OpenWork) se problem kaise solve karein, 7 Principles se jo 80% real-world use cover karte. 5 bullets 60% value dete: Action over talk, Code over prose, Verify don't trust, Small reversible steps, Files are memory.

1. P1 Bash + P2 Code as Interface

- **P1 -- Bash is the Key:** brief the hands, not just the mind -- name the input, name the output, cut 'how' verbs.
- **P2 -- Code as Universal Interface:** complex work gets done by having AI write a small program, not by repeating prose. Your 2 jobs that never change: define the problem, verify the output.

2. P3 Verification + P4 Decomposition

- **P3 -- Verification as a Core Step:** whoever produced the output is its worst verifier -- use an INDEPENDENT path. Asking the AI 'is this correct?' is not real verification.
- **P4 -- Small, Reversible Decomposition:** rule of thumb -- if undoing a change takes more than 2 minutes, it was too big. Enforce show/verify/commit/wait-for-OK after every step.

3. P5 Persistence + P6 Constraints + P7 Observability

- **P5 -- Persisting State in Files:** the rules file (CLAUDE.md/AGENTS.md) is the most important; keep it under ~60 lines after the first weeks.
- **P6 -- Constraints and Safety:** 3 universal trust levers (Scope, Connections, Approvals) + a 5-level Autonomy Ladder. Rule: no walk-away trust, no scheduling either.
- **P7 -- Observability:** watch at least one full execution on every novel task. 5 symptoms of a session off the rails: irrelevant old references, long vague responses, contradictions, repeated apology without progress, talk of unmentioned files.

4. Four-Phase Workflow

- Explore (P1+P7, read-only) -> Plan (P2+P5, the most important phase) -> Implement (P4+P3, atomic+verify) -> Commit (P7, final verification). P6 wraps all 4 phases.
- **5 Failure Patterns:** The Drift (P5), The Confident Wrong (P3), The Big Bang (P4), The Scope Creep (P6), The Black Box (P7).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Bash is the Key (P1)	Brief the agent's hands with exact input/output, not vague advice.	Agent ke 'hands' ko exact input/output se brief karo, vague advice nahi.
Code as Universal Interface (P2)	Complex tasks get done through a small written program, not repeated prose.	Complex tasks chhoti written program se hoti, repeated prose se nahi.
Verification as a Core Step (P3)	The producer of output is its worst verifier -- always use an independent check.	Output banane wala uska sabse ghatiya verifier hai -- hamesha independent check.
Autonomy Ladder (P6)	5 levels: Watching closely -> Ambient -> Walk away -> Act without asking -> Scheduled.	5 levels: Watching closely -> Ambient -> Walk away -> Act without asking -> Scheduled.
Four-Phase Workflow	Explore -> Plan -> Implement -> Commit, wrapped by Constraints (P6).	Explore -> Plan -> Implement -> Commit, Constraints (P6) se wrapped.

Quick Self-Check

Question	Answer
Why is asking the AI 'is this correct?' not real verification?	The model that made the mistake will often confirm its own mistake -- true verification needs an independent path or source.
What's the rule of thumb for whether a change was 'too big' (P4)?	If undoing it takes more than 2 minutes, it was too big -- break it into smaller reversible steps.
Why should a rules file (CLAUDE.md/AGENTS.md) stay under ~60 lines after the first weeks?	Past that it stops being a quick reference and becomes documentation nobody reads in full -- think table of contents, not encyclopedia.
What are the 5 symptoms that a session has gone 'off the rails'?	Irrelevant old references, long vague responses, contradictions, repeated apology without progress, and mentions of unmentioned files/connectors.

Where This Leads / Sources

Leads to: From One-Off to Worker. Full worked capstone + principles-per-tool matrix in 09-10.

Source: docs/problem-solving-crash-course/ (this repo), content originally via Zia Tutor AI connector.